DOCUMENTATION OF CORRECTIVE, PREVENTIVE, AND IMPROVEMENT ACTIONS FOR THE SOFTWARE

PROJECT: Traduce Señas

TEAM MEMBERS:
Luis Alejandro Duarte Aldana
Juan Mauricio Suaza Solorzano
Juan Camilo Fierro Esquivel

NATIONAL LEARNING SERVICE – SENA
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Introduction

This document records the actions that the Traduce Señas team has taken and plans to take to correct the defects found, prevent future problems, and continuously improve the software. It corresponds to the corrective, preventive, and perfective maintenance activities described in ISO/IEC 14764 (ISO, 2006).

The actions come from four sources: the code and security audit carried out during the migration to microservices, the test results, the project risk register, and the sprint retrospectives.

Definitions

Table 1
Types of action

Type	Definition	When it applies
Corrective	Removes the cause of a defect or failure that has already occurred	After an error is detected in testing, auditing, or use
Preventive	Removes the cause of a potential problem before it occurs	When an identified risk or a detected weakness exists
Improvement	Increases the quality, performance, or value of the software when no defect exists	As a result of retrospectives, metrics, or new needs

Methodology

Each finding is managed with the plan-do-check-act cycle:

1. **Identify:** record the finding with its source, date, and severity.
2. **Analyze the root cause:** ask “why?” until reaching the cause that, if removed, prevents it from recurring.
3. **Define the action:** owner, target date, and verifiable closure criterion.
4. **Execute** in a dedicated branch, with an automated test that reproduces the problem whenever possible.
5. **Verify** that the test passes and that no regressions appear (full suite green).
6. **Close** and record the lesson learned.

Table 2
Severity scale

Severity	Criterion	Target response time
Critical	Exposes data or allows unauthorized access; the system does not work	Immediate, before any deployment
High	A main function fails or there is a relevant security risk	In the current sprint
Medium	Partial failure with an available workaround	In the next sprint
Low	Cosmetic or maintainability detail	According to backlog capacity

Corrective actions

The code and security audit of phase 4 of the migration to microservices found 2 critical issues, 2 high-severity issues, and 7 defects. The following table presents the actions taken.

Table 3
Corrective actions executed

ID	Finding	Root cause	Action	Severity	Status
AC-01	The service started even if JWT_SECRET was missing or weak	Permissive default value in the configuration	JWT_SECRET required (32+ characters), HS256 with issuer and audience	Critical	Closed
AC-02	Databases and broker exposed with default passwords (guest/guest, postgres/postgres)	docker-compose intended for development only	Ports only on 127.0.0.1; passwords required from .env	Critical	Closed
AC-03	Predictable recovery code with no attempt limit	Use of a non-cryptographic generator	crypto.randomInt, attempt counter, and strict limit on /api/auth	High	Closed
AC-04	Any user could import or delete sign templates	Route without role verification	Only ADMIN (models.manage permission) can write templates	High	Closed
AC-05	Open CORS and gateway routes with no service to handle them	Configuration inherited from the backend before the split into microservices	CORS_ORIGINS allowlist; orphan routes removed	Medium	Closed
AC-06	Events without contract: invalid payloads could break consumers	Each service defined its own types	Typed event contract and validators in @traduce/shared	Medium	Closed

ID	Finding	Root cause	Action	Severity	Status
AC-07	Achievement increments were lost with simultaneous translations	Non-atomic read and write	Atomic update with INSERT … ON CONFLICT DO UPDATE	Medium	Closed
AC-08	Username not escaped in the email HTML	Direct interpolation in the template	escapeHtml function in the email templates	High	Closed
AC-09	GET /api/profile/me failed if the profile did not exist	The profile is created per event and could arrive late	Profile created on the fly if it does not exist	Medium	Under verification
AC-10	The downloader for the LSC-54 set silently skipped videos	Byte-range read error	Correction of the range calculation and test with a real sample	High	Closed
AC-11	The download stopped with HTTP 429	16 simultaneous connections exceeded the server limit	4 connections, larger blocks, and waiting on code 429	Medium	Closed

Note. Verification: the automated backend suite finished with 494 tests passed and 0 failed on October 7, 2026.

Preventive actions

Preventive actions address project risks before they turn into failures.

Table 4
Preventive actions

ID	Risk it prevents	Preventive action	Owner	Indicator
AP-01	Insufficient LSC training data (R-001)	Use the public LSC-54 set and capture our own samples with consent	Technical lead	≥ 30 examples per sign
AP-02	Model accuracy degraded without anyone noticing (R-002)	Measure accuracy per sign before publishing templates; review unrecognized signs monthly	Technical lead	Accuracy ≥ 85%; unrecognized rate < 20%
AP-03	Confusion between signs with the same handshape	Evaluate with signers who did not take part in the templates; discard the signs that collide	Team	Confusion matrix per sign
AP-04	Leakage of personal and biometric data (R-006)	Mandatory consent per sample; video processed on the device; ADMIN-only access	Team	0 samples without consent
AP-05	Single server without backup (R-007)	Daily backups and quarterly restore test	Deployment owner	Restore < 4 hours
AP-06	Regressions when integrating changes	Mandatory automated tests before each pull request; code review	All	Suite green on every integration

ID	Risk it prevents	Preventive action	Owner	Indicator
AP-07	Secrets in the repository	.env in .gitignore; .env.example without real values; review on every pull request	All	0 secrets in the history
AP-08	Vulnerable dependencies	Monthly npm audit and controlled update	Technical lead	0 critical vulnerabilities
AP-09	Role concentration in a single person (R-004)	Documentation for each service (README, runbook) and a technical onboarding guide	Team	Any team member can deploy

Improvement actions

Table 5
Improvement plan

ID	Improvement	Expected benefit	Priority	Deadline
AM-01	Structured logs with Pino and a correlation identifier across services	Much faster incident diagnosis	High	Next sprint
AM-02	/health/ready endpoint that checks databases and broker	Deployments that detect dependency failures	High	Next sprint
AM-03	Latency and error metrics with Prometheus	Measure compliance with the performance NFR (RNF)	High	Two sprints
AM-04	Persist the classified errors in error_logs	Failure trend analysis (RF10)	Medium	Two sprints
AM-05	Recognition engine that considers hand location and movement	Distinguish signs such as “good morning,” “good afternoon,” and “good night”	High	Next version
AM-06	Two-hand sign recognition	Cover most of the signs in the LSC-54 set	Medium	Next version
AM-07	Load tests with k6 in the pipeline	Detect degradation before production	Medium	Two sprints
AM-08	Continuous integration in GitHub Actions (build + tests)	No change is integrated with failing tests	High	Next sprint
AM-09	Complete missing prototypes (notifications and 2 other screens)	Full design of the 21 screens	Low	Two sprints
AM-10	Pilot test with deaf users at an educational institution	Real-world validation of specific objective 4	High	Next version

Lessons learned

Table 6
Lessons learned

Situation	Lesson
The requirements catalog reported traceability that was not real (17/17 and 77/79)	Every coverage metric must be recalculated from the source, not copied

Situation	Lesson
The “rep_N” copies in the LSC-54 set were shifted versions of the same recording	Before downloading bulk data, study its structure with a sample
Wrist normalization removes hand location	Measure how distinguishable the signs are before promising a vocabulary
A development agent stopped midway through phase 5 due to a usage limit	Split the work into small commits and keep the state documented (ESTADO_TRABAJO.md)
The project budgets had four different totals	A data point must have a single source; the others reference it

Follow-up

The status of the actions is reviewed in every sprint retrospective. An action is only closed when there is verifiable evidence: an automated test that passes, a referenced commit, or a measured indicator. Open actions are kept in the technical backlog of the documentation repository (15-project-control/technical-backlog.md).

Conclusions

The corrective actions closed the two critical and two high-severity issues found in the audit, and the test suite supports this. The preventive actions focus on training data quality and the protection of personal data, which are the highest risks of the project. The improvement plan prioritizes observability and a recognition engine that takes hand movement into account.

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